

Multi-tenant SaaS on Kubernetes

On-demand, isolated tenant instances provisioned
entirely through GitOps and Crossplane on GKE.

GROUP B

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Agenda



Platform overview

01

One diagram, one story — the whole stack at a glance



Scaling forecast

02

Today → 100 → 1000 tenants



Capacity & cost

03

Plan vs. real GCP billing, and cost-per-tenant



Live demo

04

Create a new tenant from a 7-line claim



Elective: monitoring

05

Application health & resources page

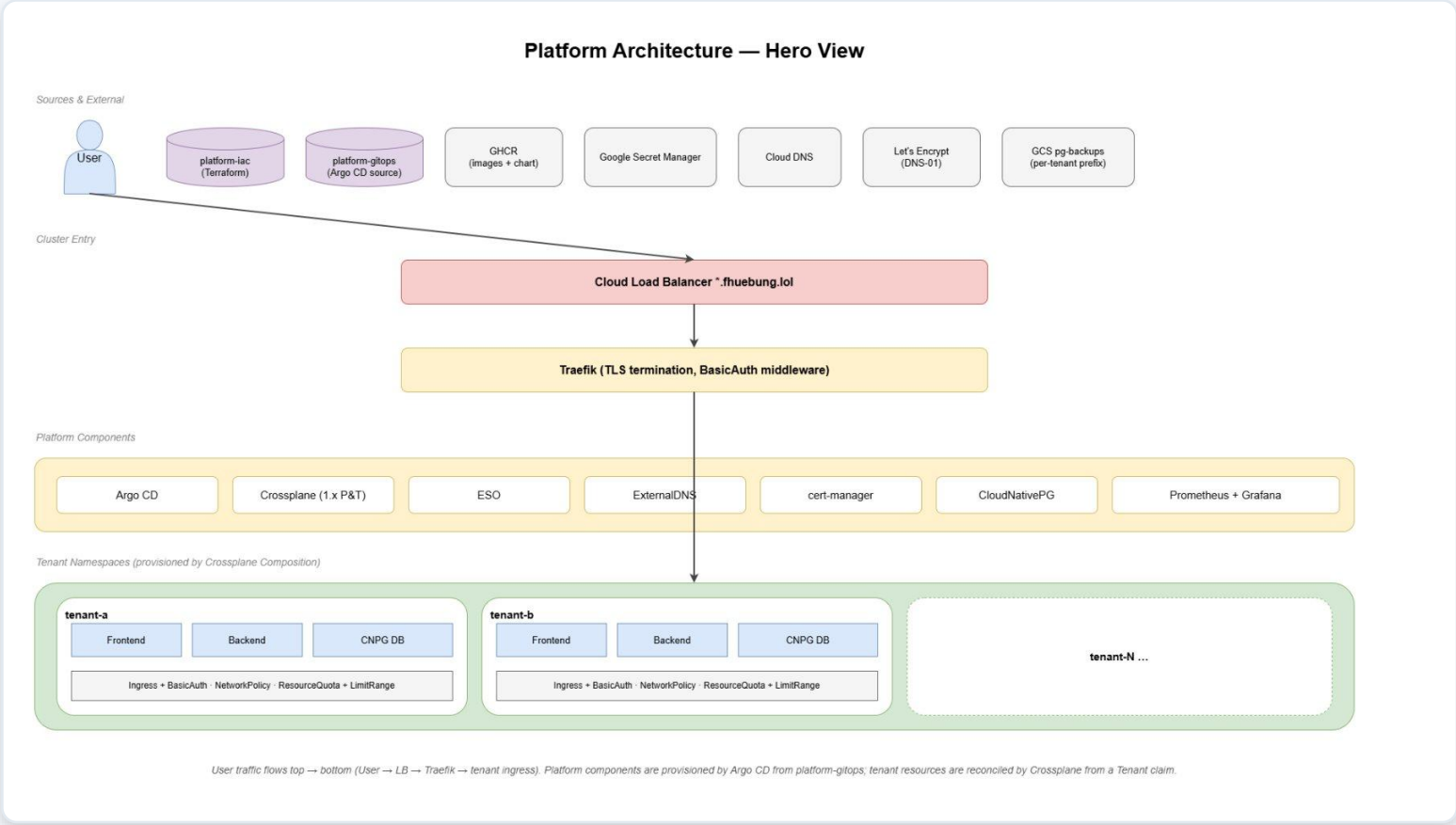


Wrap-up & Q&A

06

Where we land on the four grading pillars

Platform overview



THE STACK

 **Terraform → GKE**
IaC, zonal Standard cluster

 **Argo CD**
GitOps control plane

 **Crossplane**
P&T Compositions, tenant glue

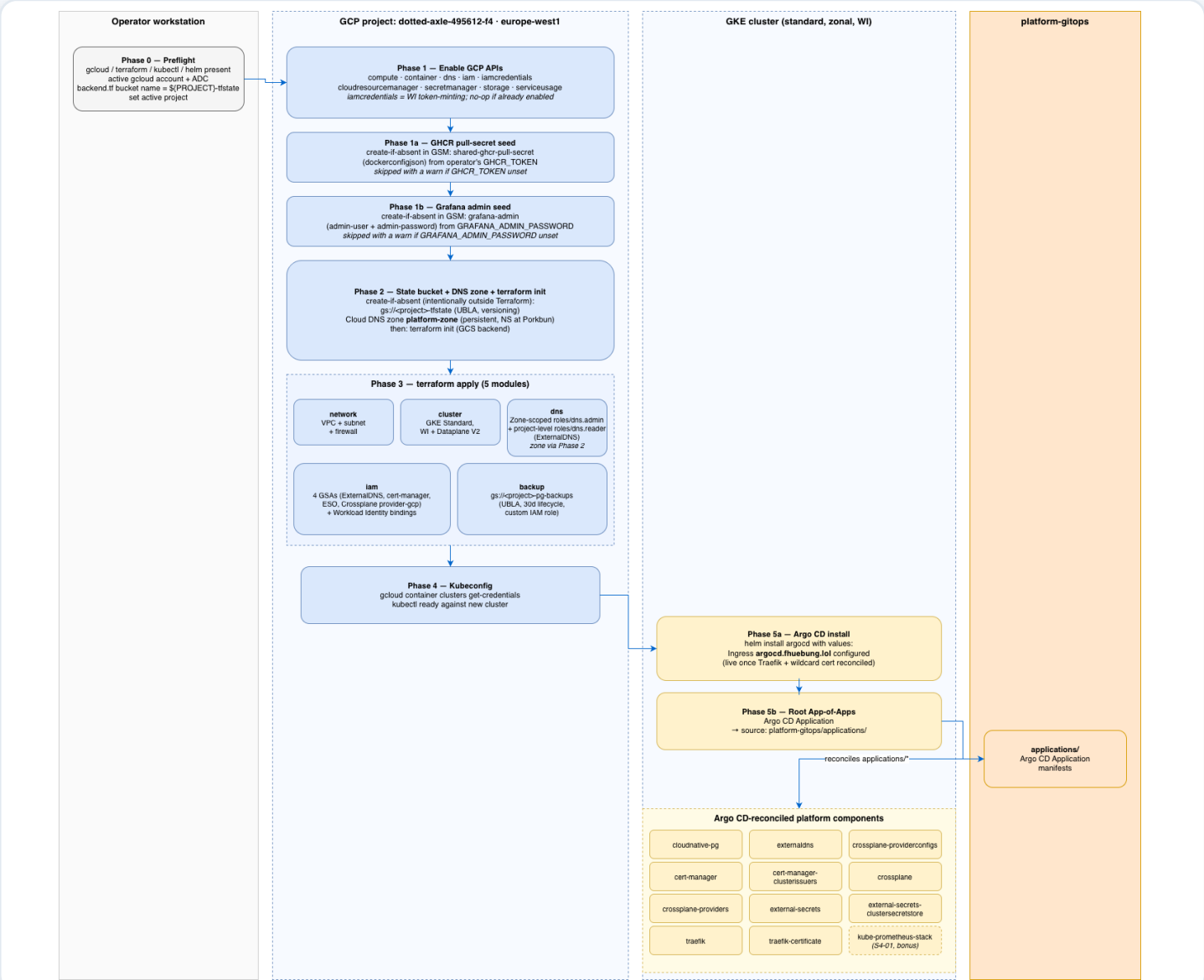
 **CloudNativePG**
Per-tenant Postgres

 **ESO + Secret Mgr**
No plaintext secrets

 **ExternalDNS + cert-mgr**
DNS-01 wildcard TLS

 **Traefik + kube-prom-stack**
Ingress + monitoring

Day 1 bootstrap

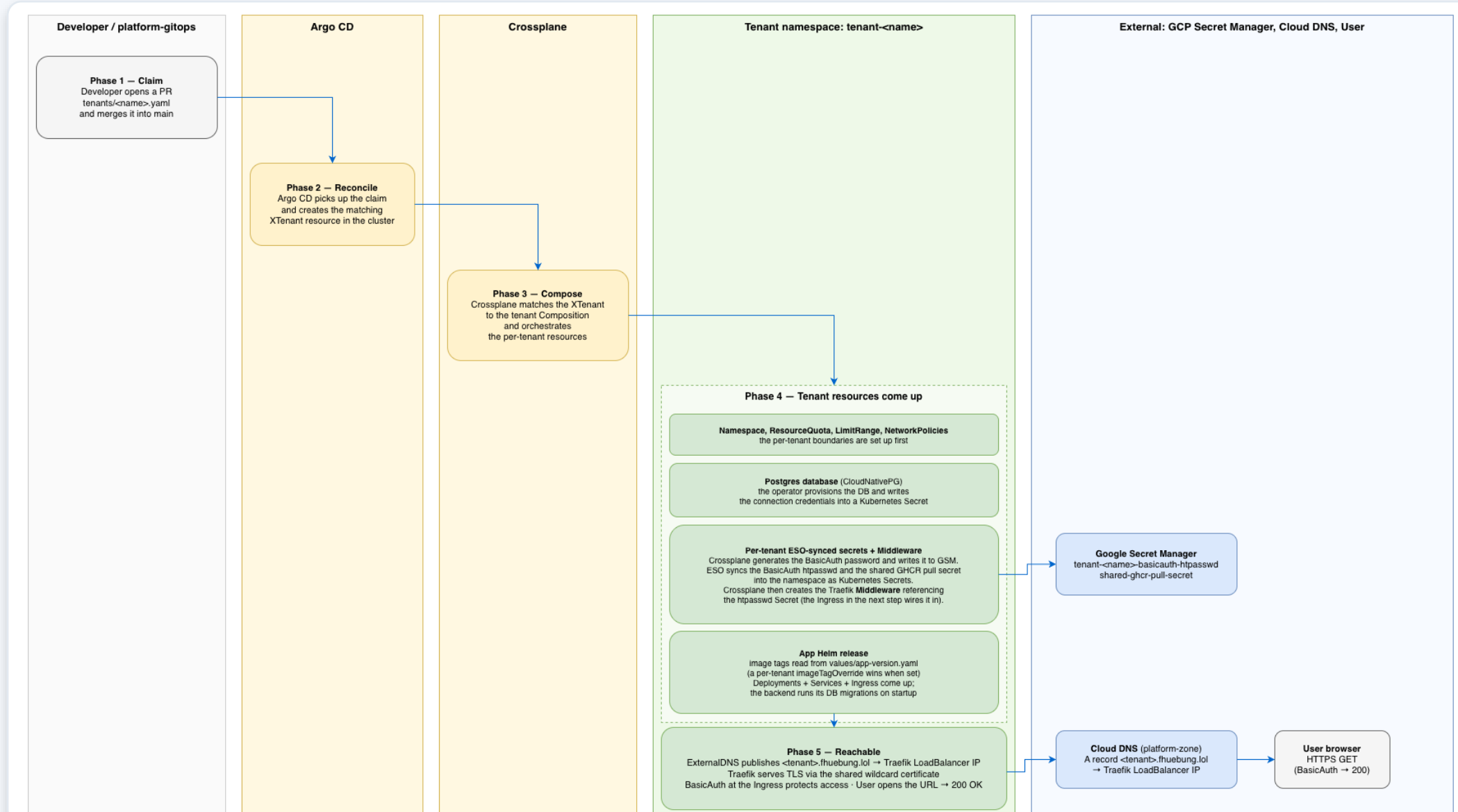


```
$ bootstrap/bootstrap.sh
```

the only documented manual step

- 0-2 Preflight, APIs, state bucket + DNS zone
- 3 terraform apply - network, cluster, IAM, dns, backup
- 4 kubeconfig against the new cluster
- 5 Argo CD install + root App-of-Apps

Behind the scenes - Day 2 provisioning



Scaling forecast

TODAY

3–6 tenants · baseline

- ✓ 3 × e2-standard-4 (12 vCPU, 48 GB)
- ✓ 1 CNPG + 1 backend + 1 frontend / tenant
- ✓ Single zonal GKE control plane
- ✓ One Composition (small / medium tier)

100 TENANTS

incremental — same architecture

- ✓ Cluster Autoscaler upper bound ~20
- ✓ ~100 namespaces × ~5 Pods, under node limits
- ✓ Still one CNPG cluster per tenant
- ✓ Logging cost scales linearly - watch it

1000 TENANTS

architectural shifts required

- ✓ Regional GKE for control-plane SLA
- ✓ Node pools by class: DB vs app
- ✓ Shard across GKE cells, or shared CNPG (small)
- ✓ Argo CD ApplicationSet sharding
- ✓ Egress + LB → next cost cliff

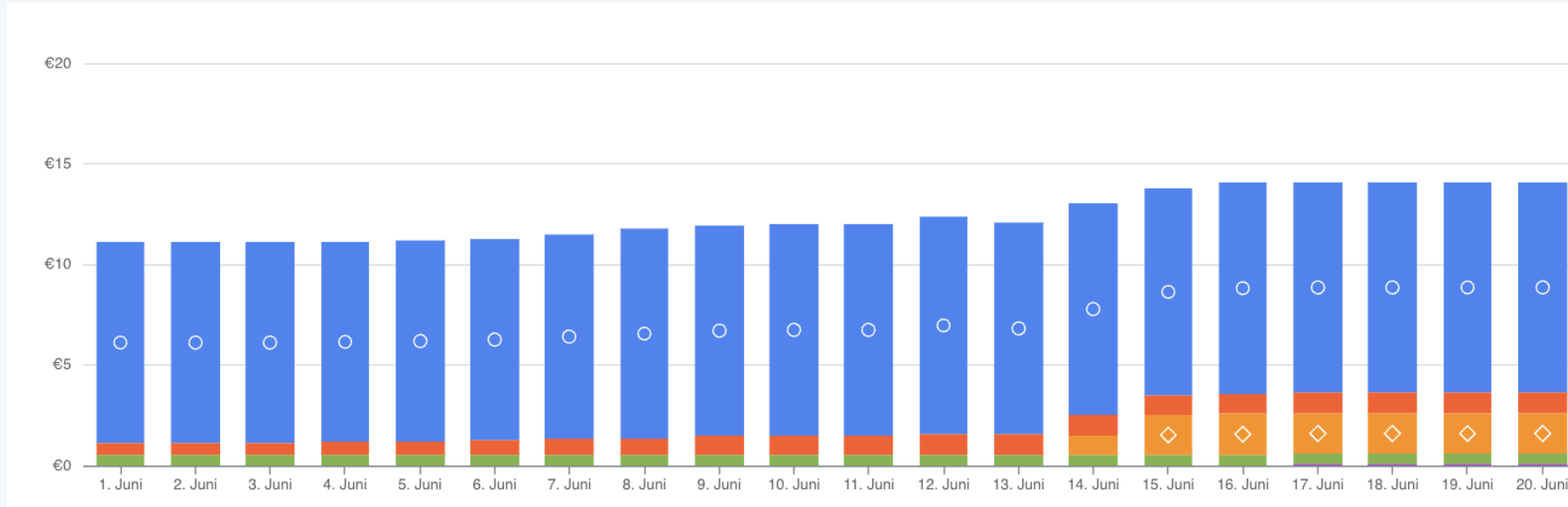
Assumption: linear extrapolation holds while tenants stay on the small tier and traffic matches the demo workload.

Costs: Held within 4% per month

Steady-state = daily rate on 2026-06-20 × 30 days (first full week without K8s “Always Free” credit)

Service	Estimate	Actual (day × 30)	Actual vs. estimate	Comment
Compute Engine: workers + disks	329.85 €	~315 €	– 5 %	10.50 €/day × 30. Cluster stayed at the 3-node minimum as modelled.
Kubernetes Engine: control-plane fee	62.42 €	~62 €	± 0 %	2.06 €/day × 30 ≈ 62 €. Matches estimate exactly. GCP credit absorbed it until 13.06.; full fee paid since 14.06. https://cloud.google.com/kubernetes-engine/pricing K8s “Always Free” credit of ~64 €/month per billing account
Cloud Monitoring (Google Managed Prometheus)	0.00 €	~29 €	NOT modelled	0.98 €/day × 30. GKE auto-enables Managed Prometheus by default - forwards cluster metrics into Cloud Monitoring. We are currently not using it (our in-cluster Prometheus + Grafana stack handles all monitoring) -> can be disabled to save further costs!
Networking: Load Balancer + NAT + egress	23.41 €	~16 €	– 33 %	0.52 €/day × 30 Demo traffic well under the 50 GiB monthly egress assumption.
Cloud Logging: cluster + application logs	22.23 €	0 €	– 100 %	0.00 €/day Stayed under the 50 GiB/month “Always Free” cap - small platform, low log volume.
Cloud DNS: managed zone + lookups	0.86 €	~0.30 €	– 65 %	0.01 €/day × 30 Only our validation traffic
Secret Manager: credentials + tokens	2.49 €	~1.20 €	– 52 %	0.04 €/day × 30. Actual secret volume below the modelled level.
Cloud Storage — Terraform state bucket	0.02 €	0 €	– 100 %	0.00 €/day 1 GiB stays inside the “Always Free” allowance
TOTAL	441.27 €	~423 €	– 4 %	

Costs: Report GCP Billing Console



Per Day:

Dienst	Nutzungskosten ?
● Compute Engine	10,50 €
■ Kubernetes Engine	2,06 €
◆ Cloud Monitoring	0,98 €
▼ Networking	0,91 €
▲ Secret Manager	0,04 €
■ Cloud DNS	0,01 €
+ Cloud Storage	0,00 €

→ 14,11€/day

Costs: Summary

Granted budget per month

500€/month

by lecturer on our request

Estimate per month

441€/month

based on submitted capacity plan

Until 13.06.

~361€/month

*with K8s "Always Free" credit of ~64
€/month per billing account*

Actual from 14.06.

~423€/month

*without K8s credits
– 4 % vs. estimate ; – 15 % vs. budget*

Net effect: estimate held within 4 % once the K8s credit ended. Budget headroom: ~77 € /month against the 500 € provided + more savings for Cloud Monitoring possible!

Live demo - create a new tenant

`tenants/livedemoNN.yaml`

```
apiVersion:
platform.fhuebung.lol/v1alpha1
kind: Tenant
metadata:
  name: livedemoNN
spec:
  name: livedemoNN
  tier: small
```



THE PATH

1

PR on platform-gitops

squash-merge the Tenant claim into main

2

Argo CD syncs

picks up the manifest, creates the XTenant

3

Crossplane reconciles

xtenant-default Composition

4

Resources come up

Namespace · NetworkPolicy · Quota · CNPG · BasicAuth

5

Helm Release

provider-helm pulls app-chart from GHCR

6

Reachable

ExternalDNS record + wildcard cert → URL responds

Application health & resource monitoring

STACK

kube-prometheus-stack

self-hosted in-cluster · grafana.fhuebung.lol via Traefik + wildcard cert

Workload Identity for GCP reads — no SA-JSON · dashboards loaded via Argo CD

WHAT WE MONITOR



Cluster

node CPU / mem / disk, kubelet pressure, autoscaler events



Platform

Argo sync, Crossplane reconcile, ESO errors, cert expiry



Tenants

per-ns CPU/mem vs Quota, CNPG repl/WAL, Traefik 5xx

How did we start & worked together?



Thank you

The cluster stays live - probe anything.



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